

Faster Approximate Linear Matroid Intersection

Tatsuya Terao (Kyoto University)

SWAT'26 @ Copenhagen

Linear Matroid Intersection

input: two $r \times n$ matrices M_1, M_2

output: maximum-size column set that is linearly independent in both M_1 and M_2

e.g.

$$M_1 = \begin{matrix} & \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$M_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Linear Matroid Intersection

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$$M_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Linear Matroid Intersection

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e.g.

$$M_1 = \begin{bmatrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{OK!}$$

$$M_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} \quad \text{NG!}$$

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$$M_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

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$$M_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

OK!

OK!

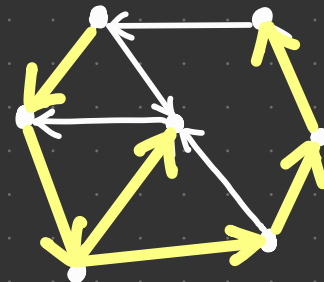
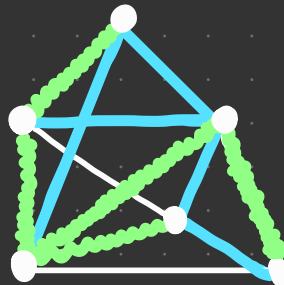
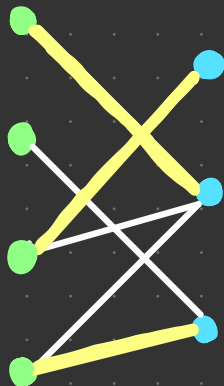
Importance of Linear Matroid Intersection

input: two $r \times n$ matrices M_1, M_2

output: maximum-size column set that is linearly independent
in both M_1 and M_2

Generalizes many important problems in Combinatorial Optimization!

e.g. Bipartite matching, Packing Spanning trees, Arborescences



Fast Algorithm for Matrices

- Product of two $n \times n$ matrices

$$\begin{matrix} n \\ \square \\ n \end{matrix} A \times \begin{matrix} n \\ \square \\ n \end{matrix} B = \begin{matrix} n \\ \square \\ n \end{matrix} ?$$

trivial : $O(n^3)$

Strassen 69 $O(n^{\log_2 7}) = O(n^{2.807})$

⋮

Alman-Duan-V. Williams
- Xu-Xu-Zhou'25

$O(n^{2.371339})$

Fast Algorithms for Matrices

- $(n \times n) \times (n \times n)$: $O(n^w)$ time ($w < 2.372$)
- Computation of rank, inv. matrix of $n \times n$ matrix: $O(n^w)$ time

Fast Algorithms for Matrices

- $(n \times n) \times (n \times n): O(n^w)$ time ($w < 2.372$)
- Computation of rank, inv. matrix of $n \times n$ matrix: $O(n^w)$ time
- $(r \times n) \times (n \times r): O(nr^{w-1})$ time

$$\begin{matrix} n \\ \boxed{A} \\ r \end{matrix} \times \begin{matrix} r \\ \boxed{B} \\ n \end{matrix} = \begin{matrix} r \\ \boxed{?} \\ r \end{matrix}$$

rectangular matrix $r \ll n$

Fast algorithm

to compute the rank of an $r \times n$ matrix

$$(n \times n) \times (n \times n) : O(n^w) \quad (w < 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

$$\bar{r} = \text{rank}(M)$$

trivial (Gaussian elimination) : $O(nr\bar{r})$

Buch-Hopcroft '74 : $O(nr^{w-1})$

Storjohann '09 : $O(nr\bar{r}^{w-2})$

Fast algorithm

to compute the rank of an $r \times n$ matrix

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

trivial (Gaussian elimination):	$O(nr\bar{r})$
Buch-Hopcroft '74 :	$O(nr^{w-1})$
Storjohann '09 :	$O(nr\bar{r}^{w-2})$

$M = r \begin{matrix} & n \\ \begin{matrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \end{matrix} \end{matrix}$

$\bar{r} = \text{rank}(M), \text{nnz}(M) := \# \text{ non-zero elem. in } M$

Cheung-Kwok-Lau '13 : $O(\underbrace{\text{nnz}(M)}_{\leq nr} + \bar{r}^w)$

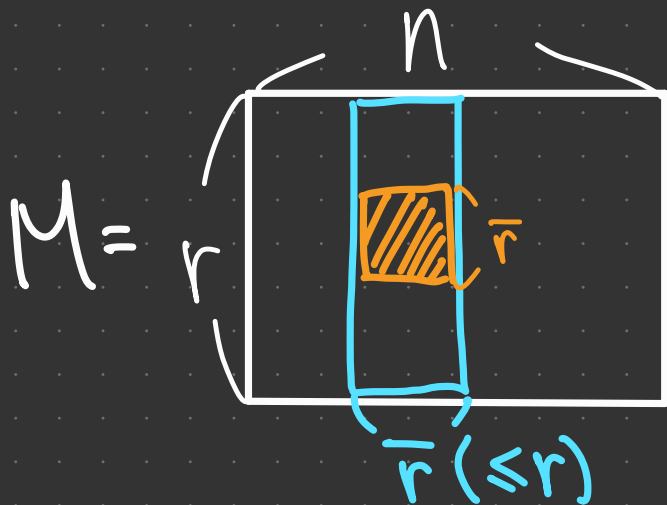
Fast algorithm

to compute the rank of an $r \times n$ matrix

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

$$\bar{r} = \text{rank}(M), \text{nnz}(M) := \# \text{ non-zero elem. in } M$$

Cheung-Kwok-Lau 13 : $O(\text{nnz}(M) + \bar{r}^w)$



input size requires to check if $\bar{r} \times \bar{r}$ matrix is full rank.

Fast algorithm

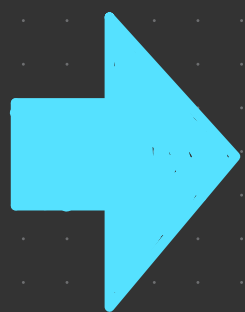
to compute the rank of an $r \times n$ matrix

$$(n \times n) \times (n \times n) : O(n^w) \quad (w < 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

$$\bar{r} = \text{rank}(M), \text{nnz}(M) := \# \text{ non-zero elem. in } M$$

Cheung-Kwok-Lau 13 : $O(\text{nnz}(M) + \bar{r}^w)$

input size requires to check
if $\bar{r} \times \bar{r}$ matrix is full-rank.



Running time that fully exploits
the sparsity of M_1, M_2 and "the smallness of $\bar{r}$ "

input: $M_1 = r$  n
 $M_2 = r$ 
 output: max. size column set
 that is linearly independent in M_1, M_2
 $r_* = \text{size of optimal sol.}$

Q. Can we solve linear matroid intersection
 in $\tilde{O}(\text{nnz}(M_1) + \text{nnz}(M_2) + r_*^w)$ time?

Time Complexities

for Linear Matroid Intersection

Edmonds '70

Cunningham '86

Gabow - Xu '89

Harvey '06

P

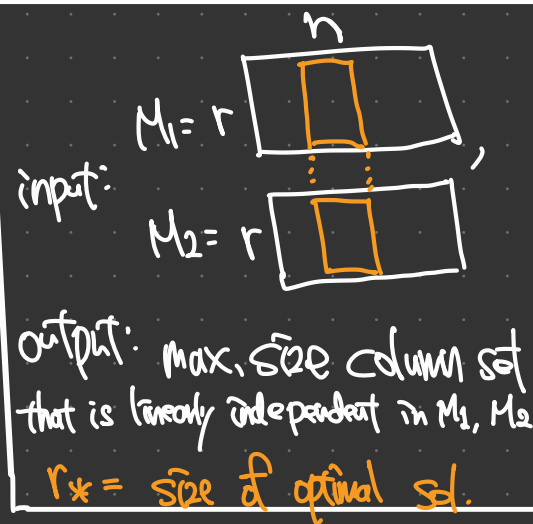
$$\tilde{O}(nr^2)$$

$$\tilde{O}(nr r_*^{1/(4-\omega)}) = \tilde{O}(nr r_*^{0.62})$$

$$\tilde{O}(nr^{\omega-1})$$

algebraic randomized algo

$$(n \times n) \times (n \times n) : O(n^\omega) (\omega < 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{\omega-1})$$



Time Complexities

for Linear Matroid Intersection

Edmonds '70

Cunningham '86

Gabow - Xu '89

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P

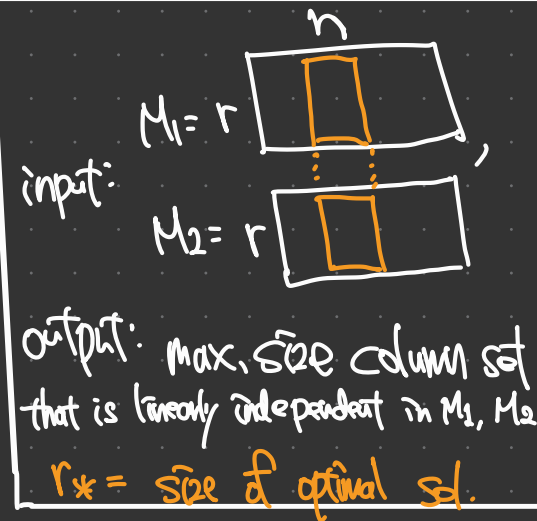
$$\tilde{O}(nr^2)$$

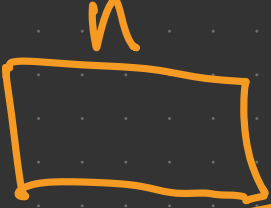
$$\tilde{O}(nr r_*^{1/(4-\omega)}) = \tilde{O}(nr r_*^{0.62})$$

$$\tilde{O}(nr^{\omega-1})$$

$$\tilde{O}(nnz(M_1) + nnz(M_2) + nr_*^{\omega-1})$$

$$(n \times n) \times (n \times n) : O(n^\omega) (\omega < 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{\omega-1})$$



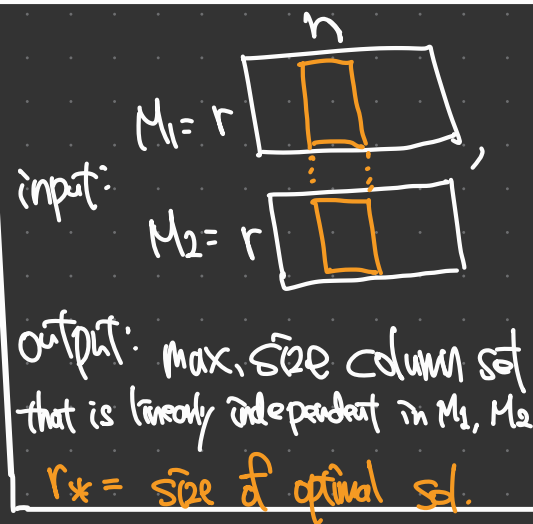
Compress M_1, M_2 into r_* 

Time Complexities

for Linear Matroid Intersection

$$(n \times n) \times (n \times n) : O(n^w) \quad (w < 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$



Edmonds '70

Cunningham '86

Gabow - Xu '89

Harvey '06

Cheng - Kank-Lau '13

This work

exact

$(1-\epsilon)$ -approx

P

$$\tilde{O}(nr^2)$$

$$\tilde{O}(nr r_*^{1/(4-w)}) = \tilde{O}(nr r_*^{0.62})$$

$$\tilde{O}(nr^{w-1})$$

$$\tilde{O}(nnz(M_1) + nnz(M_2) + nr_*^{w-1})$$

$$\tilde{O}_\epsilon(nnz(M_1) + nnz(M_2) + r_*^w)$$

Q. Can we solve linear matroid intersection
 in $\tilde{O}(\text{nnz}(M_1) + \text{nnz}(M_2) + r_*^w)$ time?

input: $M_1 = r$ 
 $M_2 = r$ 
 output: max. size column set
 that is linearly independent in M_1, M_2
 $r_* = \text{size of optimal sol.}$

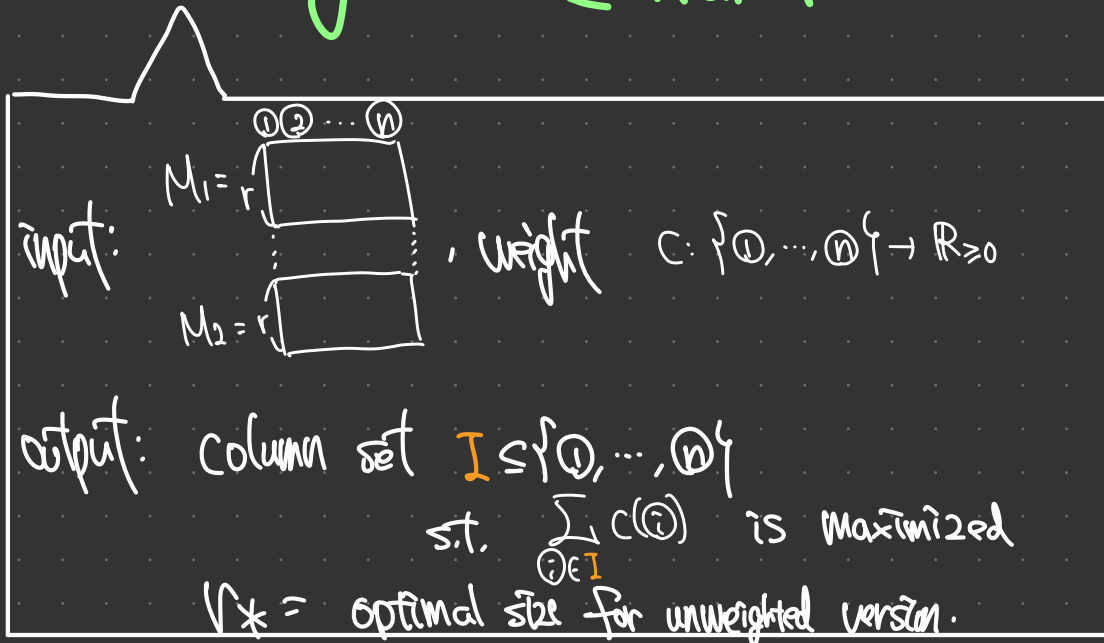
A. It's possible to obtain
 a $(1 - \varepsilon)$ -approx. sol.!

Time Complexities for

Weighted Linear Matroid Intersection

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

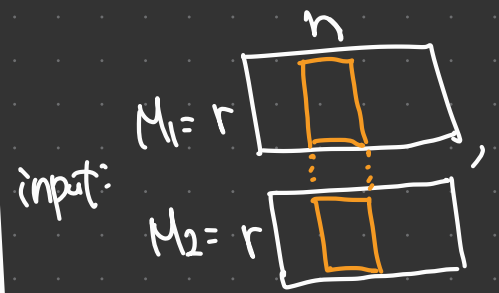


Huang-Kakimura
 - Kamiyama'16

(1- ϵ)-approx. $\tilde{O}_\epsilon(\text{nnz}(M_1) + \text{nnz}(M_2) + nr^{w-1})$

this work (1- ϵ)-approx. $\tilde{O}_\epsilon(\text{nnz}(M_1) + \text{nnz}(M_2) + r^w)$

Linear Matroid Intersection



output: max. size column set
that is linearly independent in M_1, M_2
 $r_* = \text{size of optimal set.}$

this work

$(1-\varepsilon)$ -approx.

$\tilde{O}_\varepsilon(\text{nnz}(M_1) + \text{nnz}(M_2) + r_*^w)$ time

Idea

matroid intersection algo.

in the oracle model

by Quanrud'24

+ Fast computation
of span

Matrix Intersection Algo. of Quanrud'24

input: two matrices
 $M_1 = (V, I_1), (V, I_2)$
output: max. size $I \in I_1 \cap I_2$
 $n = |V|, r_* = |I|$

① $w(e) \in [1, \dots, L]$ (for $e \in V$) $L = \tilde{\Theta}(\log(n)/\epsilon)$

② for $l = 1$ to L do

- (a) Sample $V' \subseteq V$ of size $\tilde{\Theta}(r_*)$
(each elem. $e \in V$ is sampled in proportion to $w(e)$)
- (b) Compute primal sol. $I^{(l)}$
dual sol. $(\tilde{S}^{(l)}, \tilde{T}^{(l)})$ on the sampled set V'
- (c) $S^{(l)} = \text{span}_{M_1}(\tilde{S}^{(l)})$, $T^{(l)} = \text{span}_{M_2}(\tilde{T}^{(l)})$
- (d) for $e \in S^{(l)} \cup T^{(l)}$, $w(e) \leftarrow w(e)/2$

③ Output max. size $I^{(l)}$ over all iterations

Matroid Intersection Algo. of Quanrud'24

input: two matroids
 $M_1 = (V, I_1), (V, I_2)$
output: max. size $I \in I_1 \cap I_2$
 $n = |V|, r_* = |I|$

① $w(e) \leftarrow 1$ (for $e \in V$) $L = \Theta(\log(n)/\epsilon)$

② for $l = 1$ to L do

(a) Sample $V' \subseteq V$ of size $\hat{H}_\epsilon(r_*)$
(each elem. $e \in V$ is sampled in proportion to $w(e)$)

(b) Compute primal sol. $I^{(l)}$
dual sol. $(\tilde{S}^{(l)}, \tilde{T}^{(l)})$ on the sampled set V'

(c) $S^{(l)} = \text{span}_{M_1}(\tilde{S}^{(l)}), T^{(l)} = \text{span}_{M_2}(\tilde{T}^{(l)})$

(d) for $e \in S^{(l)} \cup T^{(l)}, w(e) \leftarrow w(e)/2$

③ Output max. size $I^{(l)}$ over all iterations

linear matroid
intersection

Goal
 $\tilde{O}_\epsilon(w_R(M_1) + w_R(M_2) + r_*^\omega)$

← $\tilde{O}_\epsilon(r_*^\omega)$ time
by Haney'06

Matroid Intersection Algo. of Quanrud'24

Input: two matroids
 $M_1 = (V, I_1), (V, I_2)$
 Output: max. size $I \in I_1 \cap I_2$
 $n = |V|, r_* = |I|$

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② for $l = 1$ to L do

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 dual sol. $(\tilde{S}^{(l)}, \tilde{T}^{(l)})$ on the sampled set V'

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③ Output max. size $I^{(l)}$ over all iterations

linear matroid
 intersection

Goal
 $\tilde{O}_\epsilon(w(M_1) + w(M_2) + r_*^\omega)$

$\tilde{O}_\epsilon(r_*^\omega)$ time

by Haver'06
 How about
 this post??

Task: Fast computation of $\text{span}_m(S)$

$$\begin{aligned} (n \times n) \times (n \times n) &: O(n^w) \quad (w < 2.372) \\ (r \times n) \times (n \times r) &: O(nr^{w-1}) \end{aligned}$$

input: $r \times n$ matrix $M = r \begin{matrix} \text{---} \textcircled{1} \textcircled{2} \dots \textcircled{n} \end{matrix}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$

output: $\text{span}_u(S)$

$$= \{ i \in \{1, \dots, n\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{i\}]) \}$$



Task: Fast computation of $\text{span}_M(S)$

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

trivial
(Gaussian elimination) $O(nr^2)$

using famous fact $O(nr^{w-1})$

Requires to compute $(r \times n) \times (n \times r)$!

input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \textcircled{1} \textcircled{2} \dots \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$

output: $\text{span}_M(S)$

$= \{\textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{\textcircled{i}\}])\}$



Task: Fast computation of $\text{spann}(S)$

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

trivial
(Gaussian elimination) $O(nr^2)$

using famous fact $O(nr^{w-1})$

input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \text{ } \\ \hline \end{array} \begin{array}{c} \textcircled{1} \textcircled{2} \dots \textcircled{n} \end{array}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$
 output: $\text{spann}(S)$
 $= \{ \textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{\textcircled{i}\}]) \}$





Requires to compute $(r \times n) \times (n \times r)$!

goal $O(\text{nnz}(M) + r^w)$

Requires to avoid $(r \times n) \times (n \times r)$!

Hint: Freivalds' Algo.

$$(n \times n) \times (n \times n) = O(n^3) \text{ (w.c. 2.372)}$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n \\ \square \\ n \end{matrix} A \times \begin{matrix} n \\ \square \\ n \end{matrix} B \stackrel{?}{=} \begin{matrix} n \\ \square \\ n \end{matrix} C$$

Hint: Freivalds' Algo.

$$(n \times n) \times (n \times n) = O(n^w) \quad (w \leq 2.372)$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n & & n \\ \square & \times & \square \\ A & & B \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \square \\ C \end{matrix}$$

Time Complexity trivial

$$O(n^w)$$

Hint: Freivalds' Algo.

$$(n \times n) \times (n \times n) = O(n^w) \quad (w \in \{2, 3, 7, 2\})$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n & & n \\ \square & \times & \square \\ A & & B \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \square \\ C \end{matrix}$$

Time Complexities

trivial	$O(n^w)$
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Freivalds' A	$\hat{O}(n^2)$
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randomized algo.

Hint: Freivalds' Algo.

$$(n \times n) \times (n \times n) = O(n^w) \quad (w \leq 2.372)$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n & & n \\ \square & \times & \square \\ A & & B \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \square \\ C \end{matrix}$$

Time Complexities trivial

$$O(n^w)$$

Avoid to compute $(n \times n) \times (n \times n)$

Freivalds' A

$$O(n^2)$$

randomized algo.

Hint: Freivalds' Algo.

randomized $\tilde{O}(n^2)$ time

$$(n \times n) \times (n \times n) = O(n^w) \quad (w \leq 2.372)$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n \\ \boxed{A} \end{matrix} \times \begin{matrix} n \\ \boxed{B} \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \boxed{C} \end{matrix}$$

① Sample an n -dim. vector $r \in \{0, 1, \dots, R-1\}^n$

② if $A(Br) = Cr$, then "Yes", otherwise "No".

$$\boxed{A} \left(\boxed{B} \begin{matrix} \color{red}{\boxed{r}} \end{matrix} \right) = \boxed{C} \begin{matrix} \color{red}{\boxed{r}} \end{matrix}$$

Hint: Freivalds' Algo.

randomized $O(n^2)$ time

$(n \times n) \times (n \times n) : O(n^3)$ ($w \leq 2,372$)

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n \\ \boxed{A} \end{matrix} \times \begin{matrix} n \\ \boxed{B} \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \boxed{C} \end{matrix}$$

① Sample an n -dim. vector $r \in \{0, 1, \dots, R-1\}^n$

② if $A(Br) = Cr$, then "Yes", otherwise "No".

$$\boxed{A} \left(\boxed{B} \begin{matrix} \color{red}{r} \end{matrix} \right) = \boxed{C} \begin{matrix} \color{red}{r} \end{matrix}$$

We can compute this
in $O(n^2)$ time!

Hint: Freivalds' Algo.

randomized $\tilde{O}(n^2)$ time

$$(n \times n) \times (n \times n) = O(n^w) \quad (w \leq 2.372)$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$n \begin{array}{|c|} \hline A \\ \hline \end{array} \times n \begin{array}{|c|} \hline B \\ \hline \end{array} \stackrel{?}{=} n \begin{array}{|c|} \hline C \\ \hline \end{array}$$

① Sample an n -dim. vector $r \in \{0, 1, \dots, R-1\}^n$

② if $A(Br) = C r$, then "Yes", otherwise "No".

proof: • if $AB = C$ then algo. always output "Yes"

Hint: Freivalds' Algo.

randomized $\tilde{O}(n^2)$ time

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

input: $n \times n$ matrices A, B, C

output: check whether $A \times B = C$

$$\begin{matrix} n \\ \boxed{A} \end{matrix} \times \begin{matrix} n \\ \boxed{B} \end{matrix} \stackrel{?}{=} \begin{matrix} n \\ \boxed{C} \end{matrix}$$

① Sample an n -dim. vector $r \in \{0, 1, \dots, R-1\}^n$

② if $A(Br) = C r$, then "Yes", otherwise "No".

proof:

• if $AB = C$ then algo. always output "Yes"

• if $AB \neq C$ then algo. output "No"
w.p. $\geq 1 - \frac{1}{R}$

New!: Fast computation of $\text{span}_M(S)$

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$
$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

randomized $\tilde{O}(\text{nnz}(M) + r^w)$ time

input: $r \times n$ matrix $M = r \begin{array}{|c|c|c|c|} \hline \textcircled{1} & \textcircled{2} & \dots & \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$

output: $\text{span}_M(S)$
 $= \{\textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{\textcircled{i}\}])\}$



New!: Fast computation of $\text{span}_M(S)$

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

randomized $O(\text{nnz}(M) + r^w)$ time

$W := \text{column space of } M[S]$

input: $r \times n$ matrix $M = r \begin{matrix} \text{---} \text{---} \text{---} \text{---} \end{matrix} \begin{matrix} \textcircled{1} \textcircled{2} \dots \textcircled{n} \end{matrix}$, $S \subseteq \{1, 2, \dots, n\}$
 output: $\text{span}_M(S)$
 $= \{ \textcircled{i} \in \{1, \dots, n\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{\textcircled{i}\}]) \}$



orthogonal complement of W

① Sample a vector $vr \in W^\perp := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

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input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \textcircled{1} \textcircled{2} \dots \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{1, 2, \dots, n\}$
 output: $\text{span}(S)$
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$W := \text{column space of } M[S]$

orthogonal complement of W

① Sample a vector $vr \in W^\perp := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

• Compute a basis $\{b_1, \dots, b_k\}$ of $W^\perp$
 • $vr = \sum_i r_i b_i$ (r_i : random number)

$O(\text{nnz}(M) + r^w)$ time

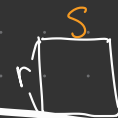
New!: Fast computation of $\text{span}_M(S)$

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randomized $O(\text{nnz}(M) + r^w)$ time

input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \textcircled{1} \textcircled{2} \dots \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{1, 2, \dots, n\}$
 output: $\text{span}_M(S)$
 $= \{i \in \{1, \dots, n\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{i\}])\}$



$W := \text{column space of } M[S]$

orthogonal complement of W

① Sample a vector $vr \in W^\perp := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

② $\text{span}_M(S) = \{i \in \{1, \dots, n\} \mid vr^T M_{\cdot i} = 0\}$

$\text{nnz}(M)$ time

New!: Fast computation of $\text{span}_M(S)$

$$(n \times n) \times (n \times n) : O(n^\omega) \quad (\omega < 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{\omega-1})$$

randomized $O(\text{nnz}(M) + r^\omega)$ time

input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \textcircled{1} \textcircled{2} \dots \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$

output: $\text{span}_M(S)$

$= \{ \textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid \text{rank}(M[S]) = \text{rank}(M[S \cup \{\textcircled{i}\}]) \}$





① Sample a vector $w \in W := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

$O(\text{nnz}(M) + r^\omega)$
time

② $\text{span}_M(S) = \{ \textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid w^T M_{\textcircled{i}} = 0 \}$

$O(\text{nnz}(M))$ time

New!: Fast computation of $\text{span}_M(S)$

$$(n \times n) \times (n \times n) : O(n^w) \quad (w \leq 2.372)$$

$$(r \times n) \times (n \times r) : O(nr^{w-1})$$

randomized $\tilde{O}(\text{nnz}(M) + r^w)$ time

input: $r \times n$ matrix $M = r \begin{array}{|c|} \hline \textcircled{1} \textcircled{2} \dots \textcircled{n} \\ \hline \end{array}$, $S \subseteq \{\textcircled{1}, \textcircled{2}, \dots, \textcircled{n}\}$

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① Sample a vector $vr \in W := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

② $\text{span}_M(S) = \{\textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid vr^T \text{Im} \textcircled{i} = 0\}$

~~proof~~ • if $\textcircled{i} \in \text{span}_M(S)$ then $vr^T \text{Im} \textcircled{i} = 0$

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① Sample a vector $w \in W := \{v \in \mathbb{F}^r \mid \forall w \in W, v \cdot w = 0\}$

② $\text{span}_M(S) = \{\textcircled{i} \in \{\textcircled{1}, \dots, \textcircled{n}\} \mid w^T M_{\textcircled{i}} = 0\}$

proof

• if $\textcircled{i} \in \text{span}_M(S)$ then $w^T M_{\textcircled{i}} = 0$

• if $\textcircled{i} \notin \text{span}_M(S)$ then $w^T M_{\textcircled{i}} \neq 0$ w.h.p.

Conclusion

input: $M_1 = r$ 
 $M_2 = r$ 
output: max. size column set
that is linearly independent in M_1, M_2
 $r^* = \text{size of optimal sol.}$

- (weighted) linear matroid intersection

$(1-\epsilon)$ -approx. $\tilde{O}_\epsilon(\text{nnz}(M_1) + \text{nnz}(M_2) + r^*{}^w)$ time

fully exploits
"the sparsity of M_1, M_2 " and
"the smallness of r^* "